A Byte-Level Multilingual News Recommender for Microcontrollers: NAS, Micro-NAS and Binarized Micro-NAS across Flash, RAM and Energy

Abstract. On-device news recommendation is private (no behavioural data leaves the device) but faces a cold-start problem and a hard memory wall: classic recommenders carry a word-embedding table of tens to hundreds of megabytes, beyond a microcontroller's flash. We present a language-agnostic *content* encoder that distils a frozen multilingual sentence-transformer into a byte-level char-CNN whose entire vocabulary is the 256 UTF-8 bytes, removing per-language tables and covering all 14 xMIND languages with one sub-2 MB model. We study three search regimes (NAS, Micro-NAS, binarized Micro-NAS) across FP32/INT8/Binary, reporting ranking quality (AUC, MRR, nDCG on MIND) against flash, RAM and an energy proxy. INT8 Micro-NAS reaches **0.610** AUC at **204 KB**, matching an NRMS word-embedding baseline (0.607) at $\sim$1/130 the size (vs. 27 MB); INT8 NAS reaches **0.647** at 790 KB. Naive binary is weak (0.52), but ReActNet-style training with distilled initialisation recovers it to **0.572** at 186 KB. Distillation adds +3.3 AUC; run-to-run std is $\leq$ 0.012; a 64-d Matryoshka prefix is nearly free; and the exported population prior lifts cold users from chance (0.500) to 0.551. Multilingual numbers are cross-lingual transfer through one content encoder (the user model uses English clicks); on-device Pi 5/STM32H7 measurements are future work. We release the pipeline and artefacts.

Keywords: Edge AI · Neural Architecture Search · Binarized Neural Networks · Multilingual News Recommendation · TinyML · Knowledge Distillation

1 Introduction

Privacy-first readers personalise *on-device*, where the user model never leaves the node. Two obstacles dominate: **cold start** (a per-user model from zero recommends poorly) and the **memory wall** (NRMS [22], NAML [20] encode titles with a word-embedding matrix, e.g. GloVe-300d [14] over 30k–100k words = 37–120 MB, which cannot fit an MCU, and multiplies per language). We pretrain offline, on MIND [26] and its multilingual extension xMIND [4], a population cold-start prior and a compact content encoder, then compress to the edge. The encoder is a **byte-level char-CNN** distilled [17] from a frozen multilingual teacher: its only symbols are the 256 UTF-8 bytes, so one model handles all 14 xMIND languages with no per-language table.

Contributions. (1) a language-agnostic byte-level content encoder that removes the embedding-table memory wall and serves 14 languages with one sub-2 MB model; (2) a systematic architecture×precision benchmark (NAS / Micro-NAS / binarized-μNAS at FP32/INT8/Binary) against flash, RAM and energy; (3) an improved binary encoder (ReActNet + Bi-Real, distilled) lifting 1-bit AUC $0.521 \rightarrow 0.572$ at 186 KB, plus multi-seed, Matryoshka and cold-start analyses and released artefacts. To our knowledge this is the first byte-level, vocabulary-free news recommender studied under a joint architecture-search and precision sweep for microcontroller-class targets; prior on-device NAS/BNN work is vision [8, 9, 15] and prior multilingual recommenders [4] are server-scale.

2 Related Work

NRMS [22], NAML [20], LSTUR [1], NPA [21] and DKN [18] set attention/knowledge title and user encoders on MIND; the task is impression-level click ranking (AUC/MRR/nDCG). Published MIND large-test AUC spans 0.65 (DKN) to 0.68 (NRMS) [26]; PLM-based models (PLM-NR [24], Fastformer [25], UNBERT/MINER) reach 0.70–0.73 [23] but wrap BERT-scale encoders (hundreds of MB), unsuitable for MCUs, so we use NRMS, the strongest lightweight title-only model, as reference. Byte/char models [2, 27] and hashing [5] drop the vocabulary table; multilingual distillation [17] aligns translations to a shared space. MCU NAS [8, 7] and binary NAS [9] plus XNOR-Net [16], Bi-Real [11] and ReActNet [10] inform the compression; we bring this to on-device multilingual news recommendation.

3 Method

Task. Impression-level click ranking: score candidates given a user's clicked history. **Encoder.** A frozen `paraphrase-multilingual-MiniLM-L12-v2` [19] gives a 384-d anchor per English title; as xMIND titles are parallel translations, every language is distilled to the *same* anchor. The student is a byte-level char-CNN: a 256-value byte embedding, depthwise-separable 1-D conv blocks (cheap, MCU-friendly), masked mean pooling and a linear head, trained with a Matryoshka [6] cosine loss so a truncated prefix stays usable. **Recommender.** News vectors feed an additive-attention user encoder over history; scores are user·candidate dot products, trained NRMS-style with one positive and K negatives (softmax cross-entropy, AdamW [12]). **Search (three arms).** One space over $\{\text{channels}, \text{depth}, \text{out\_dim}\}$, scored by distillation quality under footprint feasibility, via evolutionary search: *NAS* (FP32, loose; accuracy ceiling), *Micro-NAS* (INT8, hard STM32H7 budget), *binarized Micro-NAS* (Binary, same budget); precision is the second axis. **Precision.** STE fake-quantization: per-channel symmetric INT8 and sign×scale binary; following XNOR-Net [16] the byte embedding and first/last layers stay FP. The binary arm adds ReActNet activations (RSign, RPReLU) [10] and Bi-Real shortcuts [11] with a distilled initialisation. **Footprint/energy.** Params/MACs via `thop`; size = params×bytes/precision;

energy proxy = MACs×Horowitz-45 nm per-op [3] (FP32 MAC 4.6 pJ, INT8 0.23 pJ, 1-bit 0.0072 pJ), a *compute-only* figure that ignores memory/SRAM energy, used only for relative comparison.

4 Experimental Setup

Data. MINDsmall [26] (50k users; train 156,965 / dev 73,152 impressions; 51,282 news) and xMINDsmall [4] (14 languages, NLLB-3.3B translations [13]). xMIND supplies translated titles keyed by news id; behaviours are reused. *Every recommender and NRMS trains on English MINDsmall clicks only* (like-for-like); xMIND is used only for cross-lingual evaluation and for the distilled-init distillation, so multilingual results are cross-lingual transfer, not native-language personalisation. MINDsmall (not MINDlarge) keeps the full study (3 arms, 3 precisions, NAS, 15-language eval) to 8.2 h on one laptop; small has no public test, so we use dev and cite NRMS MINDlarge-test 0.6776 [26] as external reference. **NAS.** Evolutionary, population 30, 15 generations; candidates scored by a 2-epoch distillation on 20k titles, budget-infeasible ones rejected; Micro-NAS thresholds (256 KB / 128 KB / 2 M MACs) sit ∼8× below the H7 ceiling so they bind. **Hardware.** Training/search on one laptop: Intel Core Ultra 9 275HX (24c), RTX 5070 Laptop GPU (Blackwell, 8 GB), 32 GB, PyTorch 2.12 (CUDA 13). Edge targets: Raspberry Pi 5 (Cortex-A76, onnxruntime) and STM32H743 (Cortex-M7 @480 MHz, 2 MB/1 MB, X-CUBE-AI); their on-device latency/energy are **future work** (boards pending), so only laptop latency and the analytical proxy are reported. **Reproducibility.** Fixed seed; SHA256-pinned data; code, notebook, artefacts and a dependency lock are released.[1]

5 Results

Table 1 (arms NAS 256-4-384, Micro-NAS 64-5-384, Bin. μNAS 96-2-384) shows INT8 is effectively lossless vs. FP32 while cutting energy 7–36× and size ≈ 1.4–2×; binarization costs 5–9 AUC points but, since the embedding and first/last layers stay FP, its size gain over INT8 is modest here. Single-inference laptop latency (batch 1) is 0.34/0.19/0.10 ms on CPU (onnxruntime) for NAS/Micro-NAS/Bin. μNAS and < 0.7 ms on GPU — all far under a 50 ms budget; CPU latency tracks MACs.

Baseline. Our reproduced NRMS (256-d, 25,355-word vocabulary) has 7.08 M parameters, 91.6% in the embedding table (∼27 MB FP32; INT8 is 6.8 MB with ΔAUC < 0.001), reaching dev AUC 0.607. The byte student *matches* it at 204 KB (33× smaller than even NRMS-INT8) and *exceeds* it at 790 KB (INT8 NAS, 0.647), with no vocabulary table.

Multilingual (cross-lingual transfer). Per-language dev AUC of Micro-NAS/INT8, one byte model, no per-language table: en .604, hat .556, jpn .551, ind .537, grn .533, ron .532, tur .530, vie .523, swh .521, zho .521, som .514,

[1] Repository URL omitted for double-blind review.

Table 1. Architecture × precision (MINDsmall dev). **Bold**: best per group; †: native precision. *Arm* = search procedure (a distinct architecture); *precision* = deployment quantization; off-diagonal cells are ablations. Energy is the Horowitz proxy.

Arm	Prec.	AUC	MRR	nDCG@10	Size (KB)	Energy (μJ)
NAS	FP32†	0.633	0.326	0.375	1566.8	168.7
NAS	INT8	**0.647**	**0.338**	**0.387**	789.8	4.65
NAS	Binary	0.588	0.297	0.342	563.1	4.25
Micro-NAS	FP32	0.605	0.307	0.352	266.8	15.90
Micro-NAS	INT8†	**0.610**	**0.314**	**0.360**	203.9	2.30
Micro-NAS	Binary	0.521	0.251	0.294	185.6	2.25
Bin. μNAS	FP32	0.593	0.290	0.336	311.4	15.43
Bin. μNAS	INT8	0.601	0.295	0.344	255.7	3.28
Bin. μNAS	Binary†	**0.546**	**0.273**	**0.317**	239.5	3.23

tha .504, tam .504, fin .495, kat .491. Transfer is strongest for Latin-script / higher-resource targets and weakest for distinct-script, low-resource languages (kat, fin), tracking NLLB quality. These measure the content encoder's cross-lingual reach, not native-language personalisation.

Distillation ablation. At Micro-NAS (64-5-384, 10 epochs), initialising the encoder from the distilled student rather than training from scratch raises AUC 0.600 → 0.633 (MRR 0.293 → 0.345), so the gain comes from distillation, not capacity; we deploy the distilled-init encoder.

Binary recovery. Naive 1-bit is near chance (Micro-NAS/Binary 0.521); ReActNet + Bi-Real with distilled init lifts it to **0.572** at 186 KB, within 0.035 AUC of the FP32 baseline at two orders of magnitude less memory — a usable ultra-low-footprint option, though INT8 leads on accuracy.

Robustness, Matryoshka, cold start. Over 3 seeds Micro-NAS/INT8 is 0.624 ± 0.004 and std stays ≤ 0.012, so single-seed numbers are stable. Truncating the student embedding 384 → 64 dims barely changes cosine to the teacher (0.454 → 0.459), so an on-device 64-d embedding is nearly free. With history masked, the history-only model is at chance (0.500); ranking cold users by the exported topic prior lifts them to 0.551, confirming its value.

Trade-off. On the accuracy–footprint front, Micro-NAS/INT8 (0.610, 204 KB, 2.3 μJ) is the on-device sweet spot and NAS/INT8 the accuracy leader (0.647, 790 KB); binary is smallest but footprint-motivated only.

6 Limitations

On-device measurements are pending (Pi 5/STM32H7 boards unavailable): we report laptop latency, computed size/RAM, and the Horowitz energy proxy, which ignores SRAM/memory traffic. **Binary deployment is analytical**: INT8 deploys via ONNX→X-CUBE-AI, but the 1-bit path needs Larq/QKeras (TensorFlow; Larq is archived and its compute engine has no Windows build), and Cortex-M7 lacks POPCOUNT, so software bit-counting can erase the MAC saving — we make no binary *speed* claim. **Byte sequences** are $\sim 5\times$ longer than words, trading flash for on-MCU SRAM bandwidth (to be quantified on-device). **Cross-lingual transfer**, not native behaviour; **MINDsmall** scale; **MT quality** varies for low-resource languages.

7 Ethics and Conclusion

MIND (Microsoft Research License) and xMIND (CC-BY-NC-SA-4.0) are used for non-commercial research; no personal data is collected; translations are machine-generated. A byte-level language-agnostic encoder removes the embedding-table memory wall and serves 14 languages with one sub-2 MB model. INT8 Micro-NAS is the practical sweet spot (204 KB, 2.3 μJ, AUC 0.610); binary, recovered to 0.572 at 186 KB, is the ultra-low-footprint option, with real 1-bit ARM execution (Larq/X-CUBE-AI on Linux) and on-device measurement as the main future work. Pipeline and artefacts (ONNX encoder, cold-start prior) are released.

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